

SYNOPSIS

At 0027 on 11 December 2013, the general cargo vessel *Paula C* and the bulk carrier *Darya Gayatri* collided in the south-west lane of the Dover Strait Traffic Separation Scheme. Both vessels were damaged but there were no injuries and there was no pollution. The collision occurred as a result of *Paula C* turning into the path of *Darya Gayatri* after the action taken by the general cargo ship's officer of the watch to avoid the Belgium registered beam trawler *Raquel* had not gone as intended.

The contributing factors identified by the MAIB investigation included:

- *Paula C*'s officer of the watch did not effectively use the electronic aids available to maintain a proper lookout. After taking action to avoid the fishing vessel, he was uncertain as to the action he should take next and he lost situational awareness.
- An intervention on the radio by Dover Coastguard was timely and well-intended but, inadvertently, it almost certainly influenced *Paula C*'s officer of the watch into taking action, which resulted in *Paula C* turning towards *Darya Gayatri*.
- *Paula C*'s officer of the watch was very inexperienced and he had not yet developed sufficient competency to keep a bridge watch in the Dover Strait at night by himself.
- Although it was dark, *Paula C*'s officer of the watch was not supported by an additional lookout.
- The master's decision to allow an inexperienced officer to keep the bridge watch by himself in the Dover Strait at night was ill-judged and contrary to international requirements.

Following the accident, Carisbrooke Shipping Ltd, *Paula C*'s ship manager, adopted a more structured approach to the training and development of its junior officers. It also issued instructions to its fleet regarding the use of an additional lookout and electronic aids for collision avoidance. The Merchant Navy Training Board has started to prepare guidance for companies and seagoing officers covering junior officer development and confidence building. In addition, the Maritime and Coastguard Agency has taken action which is aimed at improving the vessel traffic services provided by Dover Coastguard.

In view of the actions already taken, no recommendations have been made.



Raquel

1.2 NARRATIVE

1.2.1 Events leading to the collision

On 10 December 2013 at 2245 UTC¹, the general cargo vessel *Paula C* was on passage in the south-west traffic lane of the Dover Strait traffic separation scheme (TSS). The vessel was following an autopilot-controlled heading of 221° at a speed over the ground (SOG)² of 11.9 knots (kts) (**Figure 1**). *Paula C* was in ballast and its destination was Poole, England. It was a dark, clear night and the visibility was good.

Paula C's master was keeping the navigational watch. As the master completed his night orders, the second officer arrived on the bridge to take over as the officer of the watch (OOV). In preparation for the watch handover, the second officer checked the settings on the port 'X-band' radar display. As he did so, he noticed a number of radar targets following the south-west traffic lane. In particular, the second officer saw a target on *Paula C*'s starboard quarter at a range of 1.9nm. From the Automatic Identification System (AIS) data³ shown on the radar display, the second officer identified the radar target as *Darya Gayatri*, a bulk carrier in ballast on passage to Baltimore, USA. He also identified that *Darya Gayatri* was making good

¹ i.e local time in the UK. The time zone kept on board *Paula C* and *Darya Gayatri* was UTC+1.

² All speeds referred to in this report are SOG unless otherwise stated.

³ The information provided by AIS is divided into: static information, including the ship's call sign and name; dynamic information, including position, course and speed over the ground and status; voyage related information, including destination, draught, and hazardous cargoes; and short safety related messages. Static and voyage related information is transmitted every 6 minutes, or on request. The reporting interval for dynamic information is dependent on a ship's speed and whether or not it is changing course. The reporting interval for a ship at a speed of between 0 and 14kts and changing course is 4 seconds.

a course over the ground (COG) of 216° at a speed of 12.4kts and was overtaking *Paula C*. The closest point of approach (CPA) between the vessels was 0.5nm in 88 minutes time.

At approximately 2300, *Paula C*'s master finished his night orders and advised the second officer to keep to the passage plan and to call him if in any doubt. The master left the bridge at 2305. By that time, *Darya Gayatri*'s heading had been altered to 227° and the bulk carrier was 79° abaft *Paula C*'s starboard beam at a distance of 1.7nm.

The second officer monitored *Paula C*'s position using the cross track error facility on the Global Positioning System (GPS) receiver. The vessel remained on track. At 2345, *Paula C* arrived at a navigational waypoint indicating a planned course alteration. Accordingly, the OOW altered the vessel's heading to follow a track of 212°.

At 0000, *Paula C*'s OOW plotted the vessel's position on the paper chart; it remained on the planned track. At 0011, he saw a vessel 20° off the starboard bow. Through binoculars, the OOW was able to see the vessel's port side light and its deck lights. He also correlated the unidentified vessel with a radar target. From the target's AIS data shown on the port radar display, the OOW saw that the vessel was at a range of 3.9nm and had a CPA of 0.1nm. The second officer did not acquire the vessel on the ARPA or use the AIS data to determine the vessel's name or status. The OOW assessed that the vessel was crossing *Paula C*'s bow from starboard to port. He also assessed that *Paula C* was the give way vessel.

The vessel ahead of *Paula C* was the Belgium registered beam trawler *Raquel*, which was towing its nets on a COG of 153° at a speed of 4.8kts (**Figure 2**). *Raquel* was displaying the appropriate lights for a power-driven vessel underway and engaged in trawling⁴; the vessel's deck lights were also switched on. The beam trawler's AIS was transmitting static and dynamic information, including its status (engaged in fishing), SOG and COG. However, no heading information was broadcast.

Raquel's skipper was on watch in the wheelhouse and he was monitoring other vessels in the area visually, by radar and by AIS. The skipper had seen *Paula C* and *Darya Gayatri* following the traffic lane and he was aware that he needed to take action in order to keep out of their way. Accordingly, at approximately 0013, with *Paula C* 3.4nm off the trawler's port bow, *Raquel*'s skipper began the first of several alterations to port, which were intended to eventually turn the fishing vessel onto a north-westerly heading (**Figure 3**). A single, broad alteration was not possible because *Raquel*'s manoeuvrability was limited by its fishing gear.

Raquel's changes in heading were not seen by *Paula C*'s OOW, who was also still unaware of the identity of the vessel ahead of him, or that it was engaged in fishing. At approximately 0018, when *Raquel* and *Paula C* were 1.82nm apart, the OOW adjusted the heading set on the autopilot to 230°. No sound signal was made and the second officer did not look over the starboard quarter to make sure that there were no vessels in close proximity. The OOW also did not use the ARPA's trial manoeuvre facility to determine the effect of the intended alteration on the CPA's of the other vessels in the area.

⁴ The International Regulations for the Prevention of Collisions at Sea 1972 (as amended) (COLREGs) require that a vessel engaged in trawling, in addition to the lights prescribed for its length, should display two all-round lights in a vertical line. The upper light is green and the lower light is white.

As *Paula C* steadied onto its new heading, the second officer noticed that the heading of *Raquel*, which was now almost directly ahead, had changed to the north-east. In response, the OOW adjusted *Paula C*'s heading further to starboard. By 0022 *Paula C*'s heading was 266° and the fishing vessel was about 30° off *Paula C*'s port bow at a distance of 1.1nm; *Darya Gayatri* was on the cargo ship's starboard beam at a distance of 0.98nm.

Over the next 2 minutes, *Paula C*'s OOW adjusted the autopilot to alter the vessel's heading to port and then to starboard. The vessel's changes in heading between 0022:06 and 0023:35 are detailed at **Table 1**.

Time	Heading (°)
0022:06	266
0022:23	263
0022:54	255
0023:04	253
0023:15	259
0023:24	273
0023:30	282
0023:35	287

Table 1: *Paula C*'s heading between 0022:06 and 0023:35

Paula C's manoeuvring was seen by *Darya Gayatri*'s OOW, who determined that the cargo ship would now pass about 2 cables ahead of his vessel. *Darya Gayatri*'s OOW was also aware that *Raquel* was ahead of him and was engaged in fishing. He was closely monitoring both vessels. *Darya Gayatri*'s OOW was the second officer and he was accompanied on the bridge by an able seaman (AB) lookout.

Paula C's movements were also seen on radar (**Figure 4**) by the duty Dover Coastguard watch officer. He called *Paula C* via very high frequency (VHF) radio channel 11 in order to clarify the OOW's intentions. The transcript of the resulting conversation is at **Table 2**.

As *Paula C* turned to starboard, the coastguard watch officer called *Darya Gayatri* via VHF radio, channel 11. The transcript of the conversation between Dover Coastguard and *Darya Gayatri* is at **Table 4**.

Time	Station	Dialogue
0024:45	Dover Coastguard	<i>This is Dover Coastguard calling the Darya Gayatri. This is Dover Coastguard calling you channel 11.</i>
0024:49	<i>Darya Gayatri</i>	<i>Dover Coastguard, Darya Gayatri, I did copy your conversation ah about he will be doing a three sixty ahhh I'll be coming to port, over.</i>
0025:03	Dover Coastguard	<i>Are you aware of the situation, you can actually see the fishing vessel ahead of you? Is that correct?</i>
0025:08	<i>Darya Gayatri</i>	<i>Yes, Dover Coastguard, I can see the fishing vessel ahead of me. She altered her course north-west of me now over.</i>
0025:21	Dover Coastguard	<i>Roger. Thank you sir. As long as you are aware. Many thanks</i>
0025:24	<i>Darya Gayatri</i>	<i>Okay, thank you.</i>

Table 4: Transcript of the VHF exchange between Dover Coastguard and *Darya Gayatri*

1.2.2 The collision

Immediately after *Darya Gayatri*'s OOW had finished talking to Dover Coastguard, he changed to hand-steering and instructed his lookout to take the helm. He then ordered the helm hard to port; no sound signal was made. By now, *Paula C* was turning through a heading of 297° at an increasing rate. *Darya Gayatri*'s OOW was not aware that the cargo ship was under helm to starboard. He assumed that *Paula C* would pass ahead of the bulk carrier before starting to manoeuvre to the north; he expected the vessels to pass starboard to starboard.

At 0026, 18 seconds after port helm was applied, *Darya Gayatri* started to turn to port. At the same time, the second officer noticed that *Paula C* was turning towards the bulk carrier. He immediately ordered the lookout to put the helm hard-to-starboard and then telephoned the master in his cabin to inform him that there was another ship "very close". The OOW also put the engine telegraph astern for several seconds. He soon returned the telegraph to full ahead after he assessed that there was no time for an astern movement to take effect.

Darya Gayatri's master arrived on the bridge just as *Paula C*'s port bridge wing collided with *Darya Gayatri*'s port anchor. At the point of contact, which was at 0026:52, *Darya Gayatri* was heading 198° at 12.9kts; *Paula C* was heading 070° at 6.2kts (**Figure 6**).

1.5.6 Master's orders

The orders written by the master during the evening of 10 December 2013 were:

10th Dec 2013 @2110hrs (UTC+1)

1. *Observe standing orders*
2. *Give wide berth to all traffic, min CPA 01'¹¹*
3. *Maintain proper lookout at all times*
4. *Course 220° (T) BA 1630*

A/C to 222°(T) in position 51°53.8'N 002°34.4E BA 1630, 2449

A/C to 204°(T) in position 51°26.6'N 001°55.0E BA 2449, 323

A/C to 225°(T) in position 51°12.8'N 002°45.0E BA 323

1. *Call master as marked on BA 323*
2. *Report to dover coast guard as marked BA 323*
3. *Monitor/ plot EGC message and Navtex message*
4. *Maintain fire patrol*
5. *Call or consult me any time*

Have a safe watch [sic]

The master noted on the vessel's chart BA 323 the positions at which the OOW was to call him. His standing orders also required that the OOW call him:

- *If in any doubt whatsoever;*
- *If the traffic conditions or the movements of other ships are causing concern;*
- *Any time a give way vessel on a constant or near constant compass bearing closes within fifteen minutes of established C.P.A [sic]*

1.6 CHANNEL NAVIGATION INFORMATION SERVICE

1.6.1 Purpose

The Channel Navigation Information Service (CNIS) was introduced in 1972 and provides a 24 hour radio and radar safety service for shipping within the Dover Strait. By collecting, recording and disseminating maritime information, the CNIS aims to provide the latest safety information to shipping in the CNIS area. CNIS is

¹¹MAIB explanatory note – 01' = 1nm

The MGN includes:

2.8 Vessels fishing within a Scheme are considered to be using the Scheme, and must comply with the general requirements set out in Rules 10(b) and (c), however, when fishing in a separation zone they may follow any course.

2.9 The requirement that vessels fishing must not impede the passage of traffic passing through a TSS, means that they must not operate in such a manner that neither they, nor their gear, seriously restricts the sea room available to other vessels within a lane, and must take early and substantial action to avoid any risk of collision developing. [sic]

2.10 Rule 10(f) places further obligations upon fishing vessels, with regard to their responsibility not to impede the passage of any vessel following a traffic lane, and fishing vessels are not relieved from this obligation in a developing situation where risk of collision may exist. When taking any action they must, however, take account of the possible manoeuvres of the vessel which is not to be impeded.

MGN 315 (M) Keeping a Safe Navigational Watch on Merchant Vessels, provides guidance for masters and officers in charge of a navigational watch, and specifies:

- *In certain circumstances of clear daylight conditions the Master may consider that the OOW may be the sole look-out.*
- *The officer of the watch should notify the master when in any doubt as to what action to take in the interests of safety.*

1.9.3 International Chamber of Shipping

The International Chamber of Shipping (ICS) Bridge Procedures Guide provides advice to the master and officers on safe navigation. Section 3.2.7 details the circumstances when the OOW should call the master. It states:

The OOW should notify the master, in accordance with standing orders or the bridge order book, when in any doubt as to what actions to take in the interests of safety

1.10 COLLISION REGULATIONS

The following COLREGS (**Annex B**) are particularly relevant to this accident and are summarised as follows:

- Rule 2 – Responsibility. This rule allows a departure from the collision prevention regulations when following the rules will not avoid immediate danger.
- Rule 5 – Lookout. This rule states that a lookout should be kept by all available means, visual and electronic, to assess the risk of collision.

- Rule 7 – Risk of Collision. This rule requires that all means possible, including radar, should be used to assess if a risk of collision exists as early as possible. Risk of collision is primarily determined by monitoring the compass bearing of an approaching vessel.
- Rule 8 – Action to Avoid Collision. This rule requires that any action taken to avoid a collision is positive, clear and made in ample time. Such action should not result in another close quarters situation.
- Rule 10 – Traffic Separation Schemes. This rule specifies the responsibilities between vessels operating in a traffic separation scheme.
- Rule 13 – Overtaking. This rule states that the overtaking vessel must keep out of the way of the vessel being overtaken.
- Rule 15 – Crossing Situation. When two power-driven vessels are crossing each other and there is risk of collision, the vessel which has the other on its own starboard side shall keep out of the way of the other and, if possible, avoid crossing ahead of the other vessel.
- Rule 16 – Action by the give-way vessel. Every vessel required to give way must take early and substantial action to keep well clear.
- Rule 17 – Action by the stand-on vessel. Where one of two vessels is to keep out of the way, the other vessel should maintain its course and speed. The stand-on vessel may take action to avoid collision as soon as it is apparent that the give-way vessel is not taking the required actions. When taking such action, a stand-on vessel should try to avoid altering course to port for a vessel on its own port side.
- Rule 34 – Manoeuvring and warning signals. Vessels in sight of one another are to warn other vessels of their intended movement by the use of sound and light signals.

1.11 SIMILAR ACCIDENTS

The number of collisions and hazardous incidents (which resulted in a HAZREP report being issued)¹² in the south-west traffic lane of the Dover Strait TSS recorded by Dover Coastguard between 2009 and 2013 is at **Table 3**. **Table 3** also shows the number of vessels that reported to CNIS on entering the area of the south-west traffic lane for which CNIS has responsibility.

¹² A HAZREP is a notification of an apparent breach of COLREGs other than rule 10. The data in the table represents both reported and unreported incidents. A reported incident is where a vessel makes a complaint about the conduct of another. An unreported incident is one in which two vessels are involved in an apparent close quarters situation where neither vessel makes a complaint about the other.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

2.2.1 Initial assessment

Paula C and *Darya Gayatri* were on passage in the south-west traffic lane of the Dover Strait TSS. At the start of his bridge watch, *Paula C*'s second officer was aware that *Darya Gayatri* was overtaking; he was expecting the bulk carrier to eventually pass down *Paula C*'s starboard side. Except for a planned course alteration at 2345, the second officer's watch had been uneventful until 0011, when he saw *Raquel* 20° off his vessel's starboard bow.

Paula C's second officer only saw the fishing vessel's port side light and its deck lights. Significantly, he did not see its green over white trawling lights at its masthead, which was possibly due to glare from the deck lights. Consequently, *Paula C*'s OOW categorised the vessel as a power-driven vessel rather than a vessel engaged in fishing.

It is evident that the OOW's categorisation of *Raquel* led him to assume that this was a crossing situation in which *Paula C* was the 'give way' vessel, and that *Raquel* was required to 'stand on' (COLREGs rules 15 and 16). In isolation, the OOW's incorrect assessment was not dangerous. The usual options of a course alteration to starboard and/or a reduction of speed were both viable. However, because the OOW did not monitor the situation closely, he did not realise that *Raquel* had already turned to the north-east when he altered *Paula C* to starboard towards a heading of 230° shortly after 0018.

Raquel had started to alter course to port at 0013. However, it is evident that over the next 5 minutes *Paula C*'s OOW did not see the trawler's heading change either visually, or by radar, ARPA or AIS. Therefore, the OOW was neither keeping a proper visual lookout, nor was he effectively using the electronic aids available. Although *Raquel*'s rate of turn was slow due to its fishing gear, given the time the fishing vessel was turning and the extent of the turn, the change of heading and/or COG would have been apparent on both the radar display and the AIS. Indeed, the fishing vessel's COG would potentially have been updated almost immediately on the AIS display.

2.2.2 The failure of the plan

It was only as *Paula C* steadied on a heading of 230°, that *Paula C*'s OOW saw that *Raquel* had altered course towards the north-east. This made no sense to the OOW. He did not understand why a power-driven vessel that had been crossing the traffic lane would manoeuvre in this way. Instead of *Raquel*'s bearing drawing to port and its CPA increasing, the CPA was still very close. The OOW's plan had not worked.

The subsequent significant fluctuations in *Paula C*'s heading between 0022:06 and 0023:35 (**Table 1**) and during the OOW's exchange with Dover Coastguard, strongly indicate that the OOW was not coping with the demands of the situation and that he did not know what to do. Therefore, when Dover Coastguard called *Paula C* on VHF radio and asked the OOW '*are you going to do a three sixty?*' the OOW quickly interpreted this question as a suggestion, which he then readily adopted as his solution to the problem. It is highly unlikely that the OOW was considering such action before Dover Coastguard asked the question. It is also unlikely that, until the VHF exchange, the OOW was aware that the vessel that he was trying to avoid was a fishing vessel.

2.2.3 Loss of situational awareness

After *Paula C*'s OOW had finished his conversation with Dover Coastguard, he immediately applied 35° of starboard helm. The OOW clearly thought that it was safe to do so, possibly because the broadcast on VHF radio of his intention to turn to starboard was not challenged. He was oblivious to the proximity of *Darya Gayatri*, which was less than 600m off his vessel's starboard beam. The closed blinds on the aft starboard side of the wheelhouse also hampered the OOW's ability to keep an affective all round lookout.

It is evident that *Paula C*'s OOW's focus was on getting his vessel clear of the fishing vessel. In the process, he did not absorb or assimilate the information provided to him, such as the inference by Dover Coastguard that he should consider the proximity of *Darya Gayatri* or, subsequently, the stated intention of the bulk carrier's OOW to turn to port (**see Table 2**). By now, the OOW was probably not referring to the radar or AIS information and had suffered a complete loss of situational awareness such that he was unaware that *Paula C* was turning into the path of the bulk carrier. Indeed, it is apparent that he remained unaware of *Darya Gayatri*'s presence until the vessels collided.

2.2.4 Actions on board *Darya Gayatri*

Darya Gayatri's OOW was a relatively experienced second officer who was aware that his vessel was overtaking *Paula C* and that he was obliged to keep clear. It is also evident from his conversation with Dover Coastguard (**Table 3**) that he was also aware that *Raquel* was engaged in fishing and had altered course to clear the traffic lane.

When *Paula C* altered course to starboard shortly after 0018, *Darya Gayatri*'s OOW saw that the cargo ship would pass ahead of the bulk carrier at a distance of about 2 cables. However, although a CPA of 2 cables was considerably less than the minimum CPA of 1nm required by the master's night orders, the OOW neither took any action to increase the CPA, nor informed the master of the situation. Consequently, the OOW denied himself the benefit of the master's knowledge and experience at an early stage.

Nonetheless, it is evident that *Darya Gayatri*'s OOW was alert to the potential dangers as the distances between the bulk carrier, *Paula C* and *Raquel* quickly reduced. He listened to the VHF conversation between *Paula C* and Dover Coastguard and then responded quickly and stated his intention to turn to port when

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. After sighting the beam trawler *Raquel* at 0011, *Paula C*'s OOW assessed the vessel to be a power-driven vessel rather than a vessel engaged in fishing. He also assessed that *Paula C* was the 'give way' vessel. [2.2.1]
2. Although *Raquel* had started to alter course at 0013 in order to clear the traffic lane and avoid impeding the safe passage of *Paula C* and *Darya Gayatri*, this alteration was not seen by *Paula C*'s OOW for over 5 minutes. [2.2.1]
3. *Paula C*'s OOW was neither keeping a proper visual lookout, nor effectively using the electronic aids available. [2.2.1]
4. Only after *Paula C* had altered course to starboard shortly after 0018 in order to avoid *Raquel*, did its OOW see that the trawler had also altered course. The OOW's plan had not worked and he was unable to cope. [2.2.2]
5. The intervention on VHF radio by the CNIS watch officer was timely, appropriate and well-intended. However, because of the language used, it unintentionally influenced the decision-making of *Paula C*'s OOW and prompted him to turn towards *Darya Gayatri*. [2.2.2, 2.6]
6. *Paula C*'s OOW suffered a complete loss of situational awareness. He was unaware of the proximity of *Darya Gayatri* until the vessels collided. [2.2.3]
7. It is apparent from the inability of *Paula C*'s OOW to make sense of *Raquel*'s actions and his total loss of situational awareness, that he was still very inexperienced and that he had not yet developed sufficient competency to keep a bridge watch in the Dover Strait at night by himself. [2.4]
8. *Paula C*'s OOW had been in charge of only 10 bridge watches and the master had only known him for about 2 weeks. Therefore, it is astonishing that the master was sufficiently confident of the OOW's abilities to entrust him with the bridge watch in the Dover Strait. [2.5.1]
9. A revision of the bridge watchkeeping routine was warranted on board *Paula C* to ensure that the inexperienced OOW was supported by the master or the chief officer. Instead, the master treated the passage through the Dover Strait no differently than a passage in open water. [2.5.1]
10. Although it was dark, *Paula C*'s OOW was the sole lookout. An additional lookout was rarely, if ever, employed on board the vessel. [2.5.2]
11. Although the newly qualified OOW on board *Paula C* spent 3 months watchkeeping on board before being promoted to second officer, during this period he was not given the conn, which did not prepare him effectively to stand a watch alone. Therefore, his time on board was probably little different to the time he had spent understudying bridge watchkeepers during his cadetship. [2.7]

12. *Paula C's* ship manager's training and development of its junior officers warrants a more structured approach. [2.7]
13. Neither *Paula C's* nor *Darya Gayatri's* OOW called their masters as they had been instructed. Had they done so, it is highly likely that they would have intervened and the collision between *Paula C* and *Darya Gayatri* would have been avoided. [2.2.4, 2.8.1]

3.2 OTHER SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. *Paula C's* OOW did not use AIS to its full potential. As OOWs use AIS for collision avoidance to varying degrees, there is an onus on ship managers and masters to ensure that their bridge watchkeepers are fully aware of both the capabilities and limitations of the system. [2.1, 2.8.2]
2. Neither *Paula C's* nor *Darya Gayatri's* OOW made any sound signals when manoeuvring when their vessels were in close proximity. [2.8.3]